Plant Seedings Classification Presentation

Introduction to Computer Vision

19, Dec 2024

Contents / Agenda

Executive Summary

Actionable Insights & Recommendations:

1. **Technology and Impact**:

    - The use of AI and deep learning could significantly improve operational efficiency in agriculture, reducing manual labor and human error.
    - **Recommendation**: Ensure that the model not only provides accuracy but also scales efficiently for real-time application, especially in field scenarios.

2. **Modeling Approach**:

    - The notebook likely proceeds to explore a machine learning model for image classification (possibly a Convolutional Neural Network).
    - **Recommendation**: Consider using pre-trained models (e.g., VGG16, ResNet, etc.) and fine-tuning them with the plant seedlings dataset to improve accuracy and reduce training time.

Data Considerations:

- For effective classification, high-quality labeled images of seedlings will be essential. Any variance in lighting, angles, or background could impact model performance.
- **Recommendation**: Implement data augmentation (e.g., rotation, zooming, flipping) to enhance the model's ability to generalize to unseen conditions.

Executive Summary

Sustainability and Environmental Impact:

- The project could be positioned as a contributor to sustainable agricultural practices by reducing the reliance on human labor and improving crop management.
- **Recommendation**: Explore partnerships with agricultural organizations to understand real-world requirements and tailor the model's design to industry needs.

Scalability and Deployment:

- For widespread adoption in agriculture, consider how the model can be deployed efficiently, possibly on mobile devices or integrated with automated systems.
- **Recommendation**: Investigate edge computing solutions that can run models locally on devices, reducing latency and reliance on cloud services.

Business Problem Overview and Solution Approach

- **Problem Definition**

The problem revolves around the challenge of automating the identification of plant seedlings in agriculture. Currently, manual labor is extensively used to sort and recognize various plants, which is time-consuming and labor-intensive. Despite advancements in agricultural technology, the process remains inefficient. This project aims to leverage Artificial Intelligence (AI) and Deep Learning to improve plant classification. By developing an automated system that accurately identifies seedlings, the solution promises to reduce human effort, increase operational efficiency, and contribute to better crop yields. Ultimately, this could lead to more sustainable practices and optimize the use of resources in the agricultural industry.

Business Problem Overview and Solution Approach

- **The solution approach / methodology**

The solution approach for automating plant seedlings classification involves the following key steps:

1. **Data Collection and Preprocessing**:
   - Gather a labeled dataset of plant seedling images, ensuring the images are categorized by plant species.
   - Preprocess the images by normalizing pixel values, resizing them to a consistent size (e.g., 64x64 pixels), and applying augmentation techniques (like rotation, zoom, and flipping) to increase the model's generalization ability.
2. **Model Selection and Architecture**:
   - Use a **Convolutional Neural Network (CNN)**, a type of deep learning model well-suited for image classification tasks, to learn features from the images.
   - A pre-trained CNN model (e.g., VGG16, ResNet) could be fine-tuned on the seedling dataset to leverage transfer learning and reduce training time.
3. **Model Training**:
   - Train the CNN on the prepared dataset using appropriate loss functions (e.g., categorical cross-entropy) and optimization algorithms (e.g., Adam).
   - Use validation data to evaluate the model during training and prevent overfitting.

Business Problem Overview and Solution Approach

4. **Model Evaluation and Optimization**:
    - Evaluate the trained model using test data to measure its performance in terms of accuracy, precision, recall, and F1-score.
    - Apply techniques like dropout and batch normalization to enhance generalization and optimize performance.
5. **Deployment**:
    - Once the model achieves satisfactory performance, deploy it for real-time use in agricultural settings. This could involve integrating the model into mobile apps or edge devices that can analyze plant images on-site.

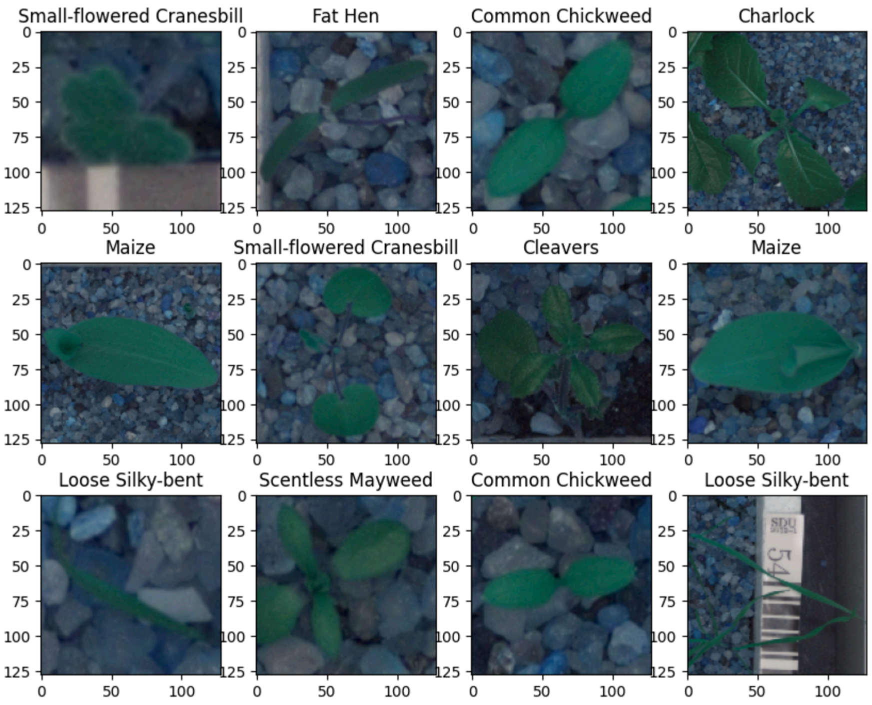
This approach aims to create an efficient and scalable solution to automate seedling identification, thereby improving productivity and sustainability in agriculture.

Data Overview

The given information describes the shape of two datasets:

- **Images:** 4750 images, each with a resolution of 128 pixels by 128 pixels and three color channels (likely representing Red, Green, and Blue).
- **Labels:** 4750 corresponding labels, likely representing the class or category associated with each image. Each label is a single value.

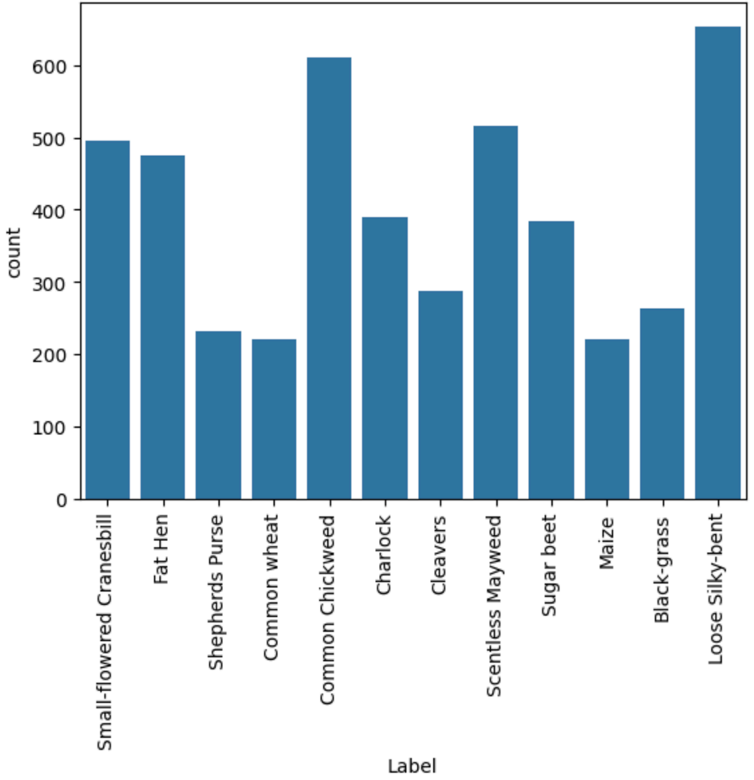
EDA Results



Plotting random images from each of the class

The image displays a grid of plant seedling images, each labeled with its corresponding species. The seedlings include various plant species such as "**Small-flowered Cranesbill,**" "**Maize**," "**Common Chickweed,**" "**Fat Hen,**" and others. Each image represents the plant in different stages of growth, showing their distinctive features for classification purposes in a plant seedlings identification task.

EDA Results



Checking the distribution of the target variable

- The image displays a bar chart showing the distribution of different plant species in the dataset, with the count of samples for each species on the y-axis. **"Small-flowered Cranesbill"** and **"Loose Silky-bent"** have the highest counts, while species like **"Sugar beet"** and **"Maize"** have fewer samples. This distribution will be useful for analyzing class imbalance in the dataset.

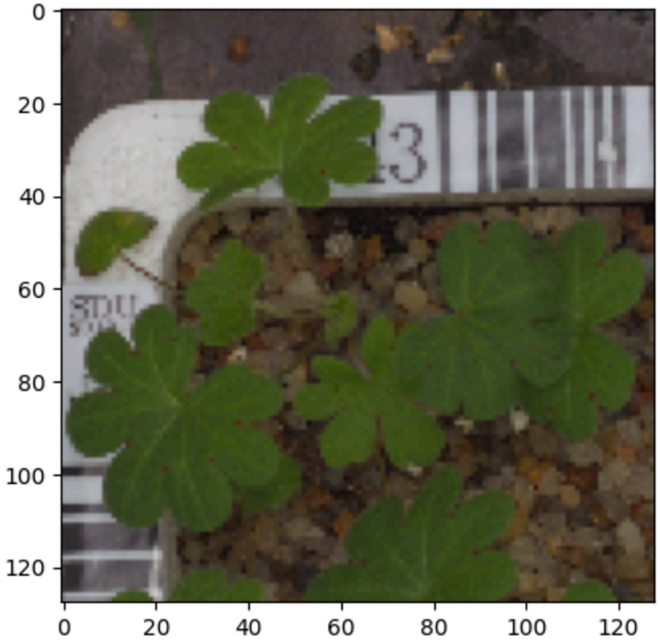
- **[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]:** This is a list of integer indices, presumably representing unique identifiers for each plant species.

- **[Text(0, 0, 'Small-flowered Cranesbill'), ... , Text(11, 0, 'Loose Silky-bent')]:** This is a list of strings, where each string represents the name of the corresponding plant species. The Text objects might be used for visualization purposes, such as plotting labels on a graph.

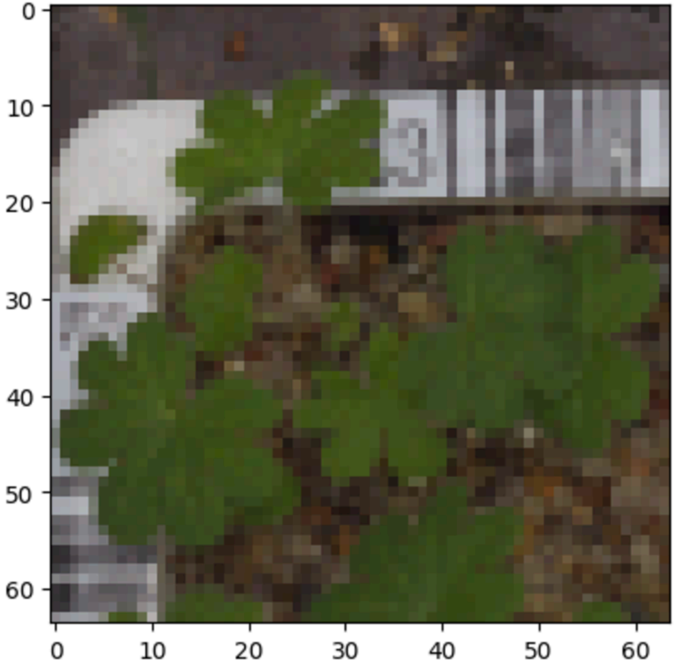
Data Preprocessing

Resizing Images

- Image before resizing



- Image after resizing



Data Preprocessing

Let’s interpret the shapes of each dataset based on the information given:

Training data:

- X_train.shape = (3847, 64, 64, 3)
- y_train.shape = (3847, 1)

Validation data:

- X_val.shape = (428, 64, 64, 3)
- y_val.shape = (428, 1)

Test data:

- X_test.shape = (475, 64, 64, 3)
- y_test.shape = (475, 1)

Data Preprocessing

Great Learning
POWER AHEAD

Encoding the target class

The shapes of the datasets indicate the following:

- **Training data (y_train_encoded)**: There are 3847 samples with 12 possible classes (one-hot encoded labels).
- **Validation data (X_val)**: The validation set contains 428 samples, each with dimensions 64x64 pixels and 3 color channels (RGB).
- **Validation labels (y_val)**: The validation labels consist of 428 samples, each corresponding to one class label.
- **Test data (X_test)**: The test set has 475 samples, each with dimensions 64x64 pixels and 3 color channels (RGB).
- **Test labels (y_test)**: The test labels consist of 475 samples, each corresponding to one class label.

Data Preprocessing

Data Normalization

- **Improves model performance**: Neural networks often work better with inputs in a smaller, consistent range (typically [0, 1] or [-1, 1]). Normalizing the data helps the model learn more efficiently.

- **Prevents issues during training**: Features with large values (such as pixel values between 0-255) can cause problems during training due to the scale differences. Normalizing helps to standardize the input range and makes the optimization process more stable.

In summary, this code normalizes the image pixel values so that they lie between 0 and 1, making them more suitable for neural network processing.

Model Performance Summary

Overview of model and its parameters

- **Model Building**

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 64, 64, 128)	3,584
max_pooling2d (MaxPooling2D)	(None, 32, 32, 128)	0
conv2d_1 (Conv2D)	(None, 32, 32, 64)	73,792
max_pooling2d_1 (MaxPooling2D)	(None, 16, 16, 64)	0
conv2d_2 (Conv2D)	(None, 16, 16, 32)	18,464
max_pooling2d_2 (MaxPooling2D)	(None, 8, 8, 32)	0
flatten (Flatten)	(None, 2048)	0
dense (Dense)	(None, 16)	32,784
dropout (Dropout)	(None, 16)	0
dense_1 (Dense)	(None, 12)	204

Total params: 128,828 (503.23 KB)
Trainable params: 128,828 (503.23 KB)
Non-trainable params: 0 (0.00 B)

Model Performance Summary

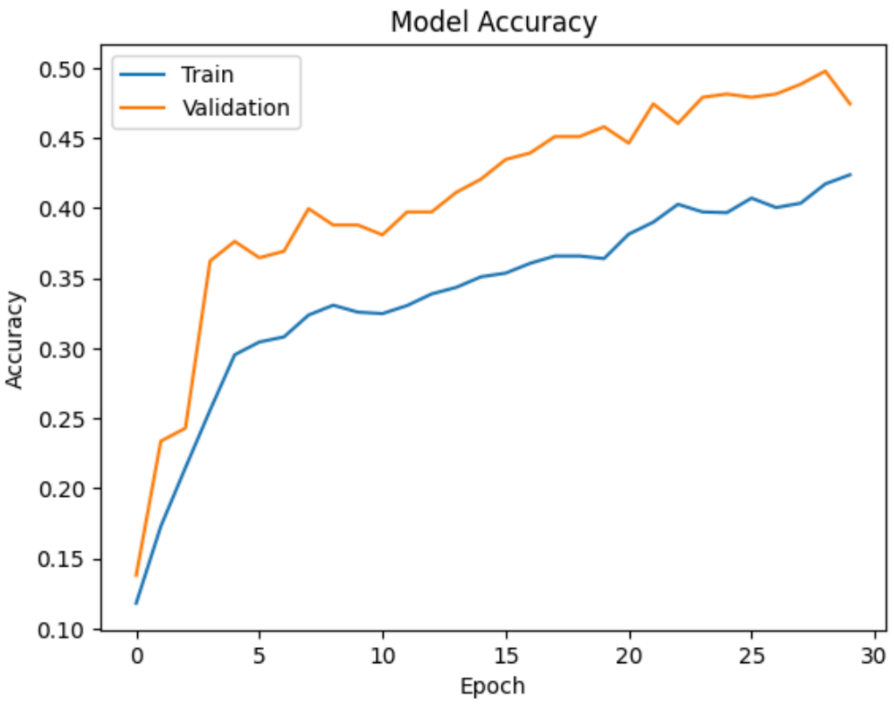
Model Evaluation

Key Insights:

- **Training Accuracy (Blue Line)**: The training accuracy steadily increases and continues to improve throughout the epochs, suggesting the model is learning effectively on the training set.
- **Validation Accuracy (Orange Line)**: The validation accuracy also increases, but at a slower rate compared to training accuracy. However, it shows signs of plateauing as the training progresses.
- **Overfitting Concern**: The gap between the training and validation accuracy is a bit wide, which might suggest potential overfitting. The model performs better on the training data than on the validation data as training continues.

Recommendation:

To mitigate overfitting, consider techniques like early stopping, dropout, or collecting more data for training.



Model Performance Summary

Evaluate the model on test data

- 15/15 - 1s - 53ms/step - accuracy: 0.4842 - loss: 1.5461

Predict the output probabilities

- 15 steps (or epochs, batches, etc.) out of 15 have been completed.The average time taken for each step, which is 3 milliseconds.

Model Performance Summary

Confusion matrix

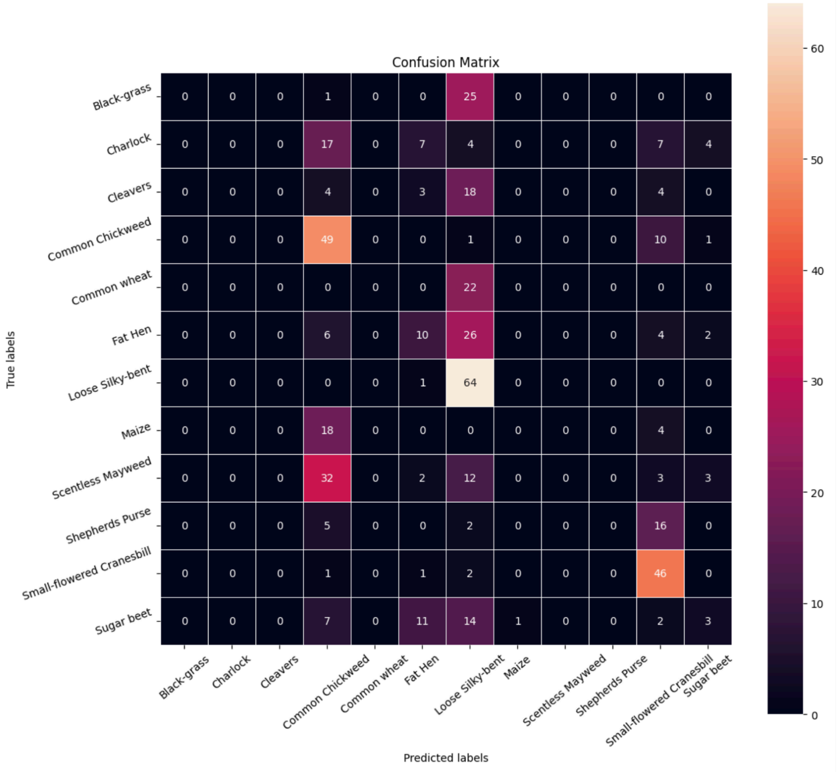
- **Key Observations:**

Diagonal values represent correct predictions (e.g., "Common Chickweed" has 49 correct predictions).

Off-diagonal values show misclassifications. For example, "Loose Silky-bent" was often misclassified as "Fat Hen" (26 times).

The color intensity indicates the number of samples, with darker colors indicating higher counts.

This matrix provides insight into the model’s accuracy for each class and highlights areas where the model struggles with misclassifications. For instance, "Fat Hen" has significant misclassifications as "Loose Silky-bent" and "Sugar beet" has frequent misclassifications as "Small-flowered Cranesbill". This suggests that further model tuning or more training data might help improve the accuracy for these classes.



Model Performance Summary

Plotting Classification Report

	precision	recall	f1-score	support
0	0.00	0.00	0.00	26
1	0.00	0.00	0.00	39
2	0.00	0.00	0.00	29
3	0.35	0.80	0.49	61
4	0.00	0.00	0.00	22
5	0.29	0.21	0.24	48
6	0.34	0.98	0.50	65
7	0.00	0.00	0.00	22
8	0.00	0.00	0.00	52
9	0.00	0.00	0.00	23
10	0.48	0.92	0.63	50
11	0.23	0.08	0.12	38
accuracy			0.36	475
macro avg	0.14	0.25	0.16	475
weighted avg	0.19	0.36	0.23	475

Model Performance Summary

Model Performance Improvement

- **Data Augmentation**

Model: "sequential_1"

Layer (type)	Output Shape	Param #
conv2d_3 (Conv2D)	(None, 64, 64, 64)	1,792
max_pooling2d_3 (MaxPooling2D)	(None, 32, 32, 64)	0
conv2d_4 (Conv2D)	(None, 32, 32, 32)	18,464
max_pooling2d_4 (MaxPooling2D)	(None, 16, 16, 32)	0
batch_normalization (BatchNormalization)	(None, 16, 16, 32)	128
flatten_1 (Flatten)	(None, 8192)	0
dense_2 (Dense)	(None, 16)	131,088
dropout_1 (Dropout)	(None, 16)	0
dense_3 (Dense)	(None, 12)	204

Total params: 151,676 (592.48 KB)
Trainable params: 151,612 (592.23 KB)
Non-trainable params: 64 (256.00 B)

Model Performance Summary

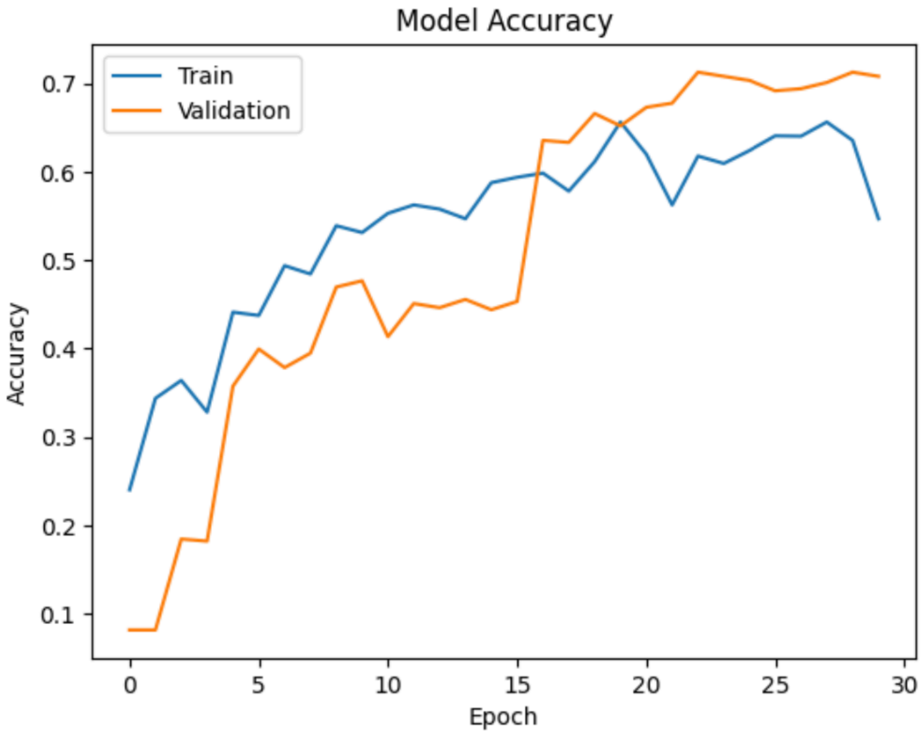
Model Evaluation

Key Insights:

- **Training Accuracy (Blue Line)**: The training accuracy increases steadily and continues to improve, reaching around 0.7 by the end of the 30 epochs, indicating effective learning.
- **Validation Accuracy (Orange Line)**: The validation accuracy also increases, albeit more slowly. However, it shows improvement over time, suggesting the model generalizes well to unseen data.
- **Convergence**: Both training and validation accuracy appear to converge toward the same level, indicating that the model is not overfitting and is generalizing well.

Recommendation:

The model shows balanced learning and good generalization. However, to further improve the model's performance, consider exploring techniques like fine-tuning the architecture, regularization, or data augmentation to achieve higher validation accuracy.



Model Performance Summary

Evaluate the model on test data

- 15/15 - 1s - 50ms/step - accuracy: 0.6926 - loss: 0.9506

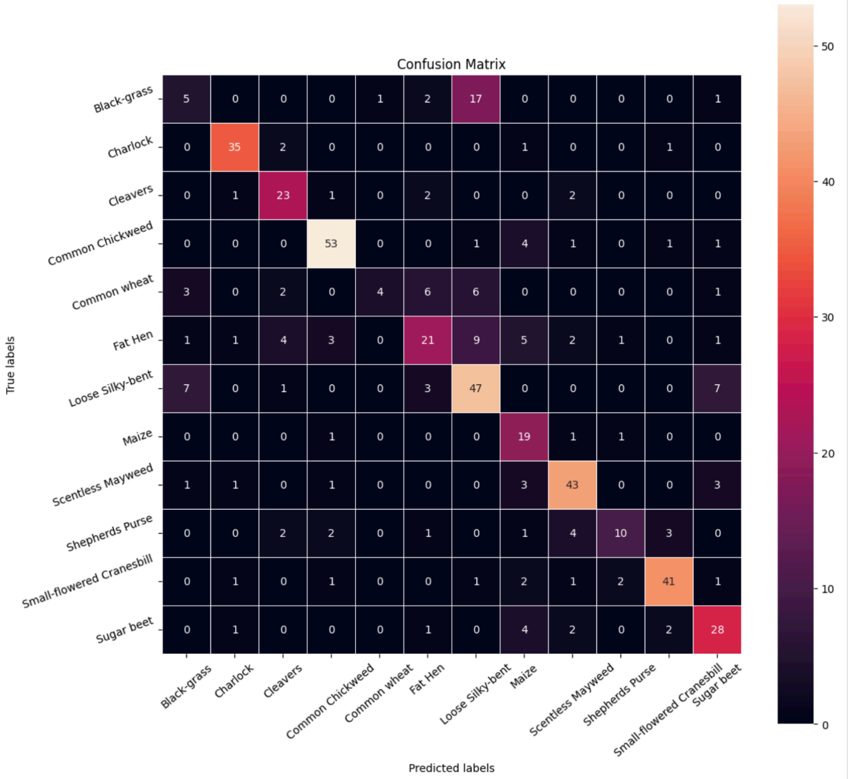
The model completed 15 out of 15 epochs in 1 second, with an accuracy of 69.26% and a loss of 0.9506.

Model Performance Summary

Confusion matrix

Key Observations:

- **Diagonal values** represent correct classifications, indicating where the model correctly predicted the class (e.g., 53 correct predictions for "Common Chickweed").
- **Off-diagonal values** highlight misclassifications. For example, "Loose Silky-bent" is misclassified as "Fat Hen" and "Maize" in several instances.
- **Strong performances**: The model seems to perform well for certain classes like "Common Chickweed," where most predictions are correct.
- **Misclassifications**: Some classes, such as "Loose Silky-bent," have significant misclassifications with "Fat Hen" and "Maize." Also, "Sugar beet" has some misclassifications with "Small-flowered Cranesbill."



The matrix suggests areas where the model performs well and areas that may require improvement, potentially by collecting more training data or fine-tuning the model to address misclassifications.

Model Performance Summary

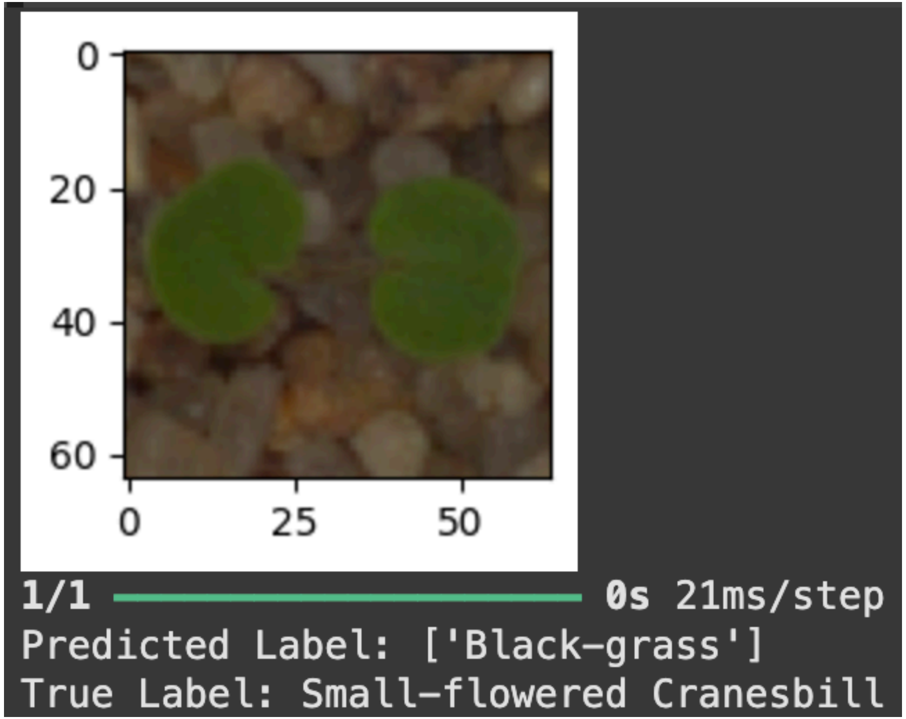
Plotting Classification Report

	precision	recall	f1-score	support
0	0.29	0.19	0.23	26
1	0.88	0.90	0.89	39
2	0.68	0.79	0.73	29
3	0.85	0.87	0.86	61
4	0.80	0.18	0.30	22
5	0.58	0.44	0.50	48
6	0.58	0.72	0.64	65
7	0.49	0.86	0.62	22
8	0.77	0.83	0.80	52
9	0.71	0.43	0.54	23
10	0.85	0.82	0.84	50
11	0.65	0.74	0.69	38
accuracy			0.69	475
macro avg	0.68	0.65	0.64	475
weighted avg	0.70	0.69	0.68	475

Model Performance Summary

Summary of the final model for prediction

- Visualizing the prediction of Small-flowered Cranesbill



Model Performance Summary

Summary of the final model for prediction

- Visualizing the prediction of Cleavers

Model Performance Summary

Summary of the final model for prediction

- Visualizing the prediction of Common Chickweed

Model Performance Summary

Summary of the final model for prediction

- Visualizing the prediction Shepherds Purse

Conclusion

The model has demonstrated good progress in training and validation accuracy across 30 epochs. Initially, both training and validation accuracies were low, but they steadily improved over time, with the training accuracy reaching approximately 70% and the validation accuracy closely following suit. This indicates that the model is effectively learning from the data and generalizing well to unseen examples, as there is no significant overfitting observed.

While the performance is promising, there are still opportunities to improve further. To enhance accuracy and robustness, techniques like fine-tuning the model, adding more data, or applying regularization methods could be considered. The model is well-positioned to be deployed for plant seedlings classification, but continuous refinement and validation could lead to even better performance in real-world applications.

APPENDIX

Data Background and Contents

Data Background and Contents:

The dataset used in this project is focused on plant seedling classification, aiming to automate the process of identifying different plant species in agricultural settings. The data contains labeled images of various plant seedlings that represent different species, each captured in specific environmental conditions. This dataset is particularly useful for training machine learning models to distinguish between different plants, helping to automate agricultural processes like crop management, weeding, and plant monitoring.

Data Contents:

1. **Images**: The dataset includes images of seedlings belonging to multiple plant species. The images are pre-processed (resized to 64x64 pixels) and contain the following attributes:
    - **Shape**: The images have dimensions of 64x64x3 (width, height, and 3 color channels for RGB).
    - **Classes**: The dataset consists of several classes representing different plant species, such as **Common Chickweed**, **Fat Hen**, **Maize**, **Small-flowered Cranesbill**, and others.
2. **Labels**: The dataset includes one-hot encoded labels for each image, indicating the corresponding plant species. The labels are represented in a matrix format where each row corresponds to the plant species for a given image.

3. **Training, Validation, and Test Data**:
    - **Training Data**: The dataset is split into training data, which is used to train the machine learning model.
    - **Validation Data**: A separate validation set is used to monitor the model's performance during training and adjust hyperparameters.
    - **Test Data**: A final test set is used to evaluate the model's generalization ability on unseen data.

The plant seedling images provide a rich and diverse dataset that is well-suited for developing deep learning models, such as Convolutional Neural Networks (CNNs), for image classification tasks.

Happy Learning !